

COURSE TITLE : ELECTRICAL AND ELECTRONIC INSTRUMENTS
COURSE CODE : 3213
COURSE CATEGORY : B
PERIODS/WEEK : 4
PERIODS/SEMESTER: 60
CREDITS : 4

TIME SCHEDULE

Module	Topics	Periods
1	Moving Coil Instruments & Multimeter	18
2	Wattmeter, Energy meter & Bridges	18
3	Cathode Ray Oscilloscope	12
4	Digital instruments & Recorders	12
TOTAL		60

COURSE OUTCOME

Module	G.O	On completion of the study of this Module the student will be able
1	1	To understand Moving Coil Instruments.
	2	To know Moving Iron Instruments.
	3	To comprehend the working of multi meter and its uses.
2	1	To know the construction and working of watt meters.
	2	To understand the use and working of energy meter
	3	To comprehend the working of impedance bridges and their uses.
3	1	To understand Cathode Ray Oscilloscope.
4	1	To comprehend Digital instruments.
	2	To know the construction and working of recorders.

On Completion of the study the student will be able

MODULE I AMMETER, VOLTMETER & MULTIMETER

1.1.0 To understand the working of Moving Coil Instruments.

1.1.1 To explain the construction and working of D'Arsonval Galvanometer.

1.1.2 To describe the deflecting, controlling and damping Torque in moving System.

1.1.3 To explain the conversion of galvanometer into voltmeter and ammeter.

1.2.0 To understand the working of Moving iron Ammeter and Voltmeter.

- 1.2.1 To describe the constructional details and the principle of working of attraction type Moving Iron Instrument.
- 1.2.2 To describe the constructional details and the principle of working of repulsion type Moving Iron Instrument.
- 1.2.3 To compare the attraction type and repulsion type moving iron instrument.
- 1.2.4 To describe the method of increasing the range of ammeters and Voltmeters.

1.3.0 To understand the working of multi meter and its uses.

- 1.3.1 To explain the block diagram of analog multimeter.
- 1.3.2 To explain the circuit in analog multimeter to measure different values of resistances.
- 1.3.3 To draw and explain the DC voltage and DC current measuring circuit in analog multimeter.
- 1.3.4 To explain the measurement techniques of AC quantities using analog multi meter.
- 1.3.5 To list the specification of a multi meter (analog type)

MODULE II WATT METER, ENERGY METER & BRIDGES

2.1.0 To understand the Watt meters & Energy meters.

- 2.1.1 To explain the different types of wattmeters
- 2.1.2 To describe with diagram the constructional details of dynamometer type Wattmeter.
- 2.1.3 To illustrate the Working principle of dynamometer type Wattmeter.

2.2.0 To understand the use of energy meter.

- 2.2.1 To describe with diagram the construction of induction type single phase energy meter.
- 2.2.2 To explain the principle of measuring energy using single phase energy meter.

2.3.0 To comprehend the working of impedance bridges and their uses.

- 2.3.1 To explain with circuit the principle of resistance measurement using DC wheatstone's bridge.
- 2.3.2 To explain with circuit the principle of impedance measurement using Maxwell's and Hay's bridges.
- 2.3.3 Describe Capacitance measurement using Wein bridge and Schering bridge.

MODULE III CATHODE RAY OSCILLOSCOPE

3.1.0 To understand Cathode Ray Oscilloscope.

- 3.1.1 To explain the Block diagram of CRO.
- 3.1.2 To brief CRT, Electron gun, Focusing, Electrostatic deflection, and Deflection sensitivity.
- 3.1.3 To explain the measurement of Voltage, Current and Time period using CRO.
- 3.1.4 To describe the basic Controls of CRO.
- 3.1.5 To illustrate Lissajous Patterns .
- 3.1.6 To explain frequency measurement using Lissajous Patterns
- 3.1.7 To describe dual beam and dual trace oscilloscopes.
- 3.1.8 To explain sampling Oscilloscope
- 3.1.9 To explain digital storage Oscilloscope
- 3.1.10 To describe different types of CRO Probes

MODULE IV DIGITAL INSTRUMENTS AND RECORDERS

4.1.0 To Comprehend Digital instruments.

- 4.1.1 To explain the block diagram of digital voltmeter.
- 4.1.2 To describe the working of Ramp Type DVM.
- 4.1.3 To describe digital multimeter with block diagram.

4.2.0 To understand the construction and working of recorders.

- 4.2.1 To describe the working of Galvanometric Recorder with figure.
- 4.2.2 To explain the working of Potentiometer type Recorder with figure.
- 4.2.3 To describe the working of X-Y Recorders with figure.
- 4.2.4 To explain the working Circular Chart Data Recorders with figure.
- 4.2.5 To describe the working of Strip Chart Recorders with figure.
- 4.2.6 To compare different types of data recorders.

CONTENT DETAILS

MODULE I

D'Arsonval Galvanometer - deflecting, controlling and damping Torque in moving System - conversion of galvanometer into voltmeter and ammeter - attraction type - repulsion type Moving Iron Instrument and Comparison - increasing the range of ammeters and Voltmeters - block diagram of analog multimeter – resistance measurement - DC voltage and DC current measuring circuit – measuring AC quantities using analog multi meter - specification of a multi meter (analog type)

MODULE II

Types of wattmeters – construction & working of dynamometer type Wattmeter - construction of induction type single phase energy meter - measurement using single phase energy meter - resistance measurement using DC wheatstone's bridge - impedance measurement using Maxwell's & Hay's bridges - Capacitance measurement using Wein bridge and Schering bridge

MODULE III

Block diagram of CRO – CRT - Electron gun – Focusing - Electrostatic deflection - Deflection sensitivity - Voltage, Current and Time period measurement using CRO - basic Controls of CRO - Lissajous Patterns - frequency measurement using Lissajous Patterns - dual beam and dual trace oscilloscopes - sampling Oscilloscope - digital storage Oscilloscope - different types of CRO Probes

MODULE IV

Block diagram of digital voltmeter - Working of Ramp Type DVM - digital multimeter block diagram - Galvanometric Recorder - Potentiometer type Recorder - X-Y Recorders - Circular Chart Data Recorders - Strip Chart Recorders - different types of data recorders.

REFERENCES

1. B L Thereja - Electrical Technology- S Chand.
2. A K Sawhney – A Course in Electrical and Electronic Measurements and Instrumentation - DhanpathRai .
3. Kalsi - Electronic Instrumentation-TMH.
4. William D Cooper- Modern Electronic Instrumentation and Measurement Techniques.- PHI.